

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed **machine learning** models using **Python** and **scikit-learn** to achieve **95%** accuracy in predictive analytics, as measured by **mean squared error**, by implementing **ensemble methods** and **hyperparameter tuning**.
- Developed **data pipelines** using **Apache Spark** and **Apache Kafka** to process **10TB** of data daily, as measured by **throughput**, by optimizing **data serialization** and **cluster configuration**.
- Improved **model performance** by **25%** using **transfer learning** and **fine-tuning** techniques, as measured by **F1 score**, by applying **domain adaptation** and **regularization**.
- Built **real-time analytics** dashboards using **Tableau** and **Power BI** to visualize **key performance indicators**, as measured by **user engagement**, by creating **interactive visualizations** and **data storytelling**.
- Enhanced **data quality** by **30%** using **data validation** and **data cleansing** techniques, as measured by **data consistency**, by implementing **data governance** and **data standardization**.
- Deployed **machine learning models** using **Docker** and **Kubernetes** to achieve **99.9%** uptime, as measured by **model availability**, by implementing **containerization** and **orchestration**.

SKILLS

Programming Languages – Python, Java, R

Data Science – scikit-learn, TensorFlow, Keras

Data Engineering – Apache Spark, Apache Kafka, Apache Hadoop

Data Visualization – Tableau, Power BI, D3.js

Machine Learning – supervised learning, unsupervised learning, deep learning

Cloud Platforms – AWS, Azure, Google Cloud

Containerization – Docker, Kubernetes

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India

01/2021

PROJECTS

Real-Time Fraud Detection Pipeline

- Designed and developed a **real-time fraud detection pipeline** using **Apache Kafka** and **Apache Spark** to process **100k** transactions per second, as measured by **throughput**, by implementing **streaming data processing** and **machine learning**.
- Improved **fraud detection accuracy** by **40%** using **ensemble methods** and **hyperparameter tuning**, as measured by **F1 score**, by applying **domain adaptation** and **regularization**.

- Deployed the pipeline using **Docker** and **Kubernetes** to achieve **99.9%** uptime, as measured by **model availability**, by implementing **containerization** and **orchestration**.

Personalized Recommendation System

- Developed a **personalized recommendation system** using **collaborative filtering** and **matrix factorization** to recommend **10k** products to **1M** users, as measured by **precision** and **recall**, by implementing **model-based** and **memory-based** approaches.
- Improved **recommendation accuracy** by **25%** using **transfer learning** and **fine-tuning** techniques, as measured by **F1 score**, by applying **domain adaptation** and **regularization**.
- Deployed the system using **AWS** and **Azure** to achieve **99.9%** uptime, as measured by **model availability**, by implementing **cloud-based** and **serverless** architecture.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)